Exam

Introduction to Database Systems – Fall 2025

General Information

Instructions:

- You have 4 hours to answer the questions in the following sections. You may have extra time if SAP has granted you special exam conditions.
- You must provide all your answers online, directly in this Exam Quiz.

Negative points: in all multiple-choice questions, wrong answers are assigned negative points so that the total score for all incorrect answers cancels out the total score for all correct answers. That is, the sum of all correct answers is 100%, while the sum of all wrong answers is -100%. For example, if you have 5 options, two are correct and three are wrong, each correct answer is worth 50% and each wrong answer is worth -33.33%. Note that the final grade for the question cannot be negative; the lowest possible grade is zero.

Important: it is your responsibility to ensure that you SUBMIT and FINISH your attempt: if an exam is not properly submitted, we WILL NOT take it into account.

Good luck!

ER Modeling, DDLs, and Normal Forms (25 points)

Question 1 (7 points)

Draw an entity-relationship (ER) diagram for the following use case. The ER diagram must follow the notation presented in the "Principles of Database Management" (PDBM) book, as amended in lectures. Make sure to include entities, relationships, primary keys, cardinalities, and other features of ER models where applicable. Add an ID as a primary key only if no other attribute can serve as a primary key. Adding unnecessary primary keys will result in a loss of points. Export and upload the diagram in PNG or JPG format.

Runners can compete in multiple races. Each runner has a first name, last name, and date of birth, and only runners aged 16 or older are eligible to compete. Races are identified by a unique race ID and include

a location (city and country) and a length. Each runner is assigned a bib number for each race in which they compete. The number of runners per race must not exceed 1,000. Medals are awarded to runners who finish a race. Each medal is identified by a serial number unique to the race and has a type (gold, silver, or bronze). Only runners who finish the race receive a medal.

Question 2 (3 points)

Which of the following rules can be represented in an ER diagram (not necessarily your diagram drawn before). The ER diagram must follow the notation presented in the "Principles of Database Management" (PDBM) book, as amended in lectures.

- ☐ Only runners aged 16+ may compete.
- ☐ A runner can enter multiple races.
- ☐ Race IDs are globally unique.
- ☐ Medals are awarded only to runners who finish a race.
- ☐ The attribute “type” of a medal must be gold, silver, or bronze.
- ☐ Serial numbers of medals are unique per race.
- ☐ A medal is awarded to not more than one runner.
- ☐ The number of runners per race must not exceed 1,000.
- ☐ Races allow multiple runners to compete.
- ☐ Runners may not finish a race.

Question 3 (7 points)

Write SQL Data Definition Language (DDL) statements to create tables for each entity and relationship (if necessary) identified in the ER model from Question 1. Specify appropriate data types for each attribute, and include primary keys and foreign keys to enforce the relationships between tables. Additionally, add constraints to enforce as many rules as possible from the problem statement of Question 1.

Note:

- The SQL DDL statements must be executable in DuckDB. Statements that do not run in DuckDB will result in a loss of points.
- Not all rules can be enforced via SQL DDL in DuckDB, as the CHECK constraint does not support subqueries.

Question 4 (2 points)

Which rules from the problem statement **cannot** be enforced using CHECK constraints in DuckDB because they would require a subquery.

- ☐ Only runners aged 16+ may compete.

- ☐ A runner can enter multiple races.
- ☐ Race IDs are globally unique.
- ☐ Medals are awarded only to runners who finish a race.
- ☐ The attribute "type" of a medal must be gold, silver, or bronze.
- ☐ Serial numbers of medals are unique per race.
- ☐ A medal is awarded to not more than one runner.
- ☐ The number of runners per race must not exceed 1,000.
- ☐ Races allow multiple runners to compete.
- ☐ Runners may not finish a race.

Question 5 (3 points)

Given a relation *Finisher(Runner, T-Shirt-Size, Race, Location(City, Country), FinalPosition)* where *Location* is a composite attribute type containing *City* and *Country*, and the following functional dependencies:

- Runner → T-Shirt-Size
- Race → City, Country
- Runner, Race → FinalPosition

What normal form is the relation in?

- Not in 1NF
- 1NF
- 2NF
- 3NF / BCNF

Question 6 (3 points)

If the relation in Question 3 is not in 3NF / BCNF, normalize the relation to the highest possible normal form (3NF / BCNF) and select the resulting relations. If the relation is in 3NF / BCNF, select the answer "The relation is already in 3NF / BCNF".

- ☐ The relation is already in 3NF / BCNF.
- ☐ FinisherInfo(<u>Runner</u>, T-Shirt-Size, <u>FinalPosition</u>)
- ☐ Location(<u>City</u>, <u>Country</u>)
- ☐ RaceInfo(<u>Race</u>, City, Country)
- ☐ RunnerInfo(<u>Runner</u>, T-Shirt-Size)
- ☐ RaceFinal(<u>Runner</u>, <u>Race</u>, FinalPosition)
- ☐ RaceLocationFinal(<u>Runner</u>, <u>Race</u>, City, Country, FinalPosition)
- ☐ FinalPosition(<u>Race</u>, <u>Runner</u>)

SQL (40 points)

You are given a dataset that you must import into a DuckDB database on your laptop. Download the file **`game_nights.sql`** from the LearnIT page and copy and paste the SQL statements into DuckDB.

The dataset contains three tables:

The **`players`** table stores information about players. It contains the following columns:

- **`first_name`**: The first name of the player.
- **`battle_name`**: The player's unique in-game alias.
- **`age`**: The age of the player.

The **`boardgames`** table stores information about board games. It contains the following columns:

- **`title`**: The name of the board game.
- **`description`**: A description of the game.
- **`max_players`**: The maximum number of players allowed for the game.
- **`rating`**: The rating of the game (out of 10).

Finally, the **`gamenights`** table stores the results and details of game nights, including which players played which games and their positions. It contains the following columns:

- **`night_number`**: A strictly monotonically increasing number for the game night.
- **`round`**: The number of the round played in a game night. There is one game played per round.
- **`date`**: The date when the game night took place.
- **`game_title`** (Foreign Key): References the **`title`** column in the boardgames table, indicating which game was played.
- **`battle_name`** (Foreign Key): References the **`battle_name`** column in the players table, indicating which player participated.
- **`player_position`**: The position (e.g., ranking) of the player in that game.

Given this dataset, answer the following questions.

Question 7 & 8 (5 points)

What is the average age of players over 30?

Enter the result and your SQL query that produces exactly this result. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

Question 9 & 10 (5 points)

How many board games have a rating above 7.5 and contain the letter "a" in their title or description?

Enter the result and your SQL query that produces exactly this result. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

Question 11 & 12 (5 points)

What is the average rating of the (unique) games played during the fourth game night (night_number = 4)?

Enter the result and your SQL query that produces exactly this result. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

Question 13 & 14 (5 points)

What is the highest number of different games played in a single game night?

Enter the result and your SQL query that produces exactly this result. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

Question 15 & 16 (5 points)

Which player(s) (by first name) have never participated in any game night?

Select the names from the multiple-choice options and enter your SQL query. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

- ☐ Barbara
- ☐ David
- ☐ Elizabeth
- ☐ James
- ☐ Jennifer
- ☐ John
- ☐ Linda
- ☐ Mary
- ☐ Michael

- ☐ Patricia
- ☐ Robert
- ☐ William

Question 17 & 18 (5 points)

Which player(s) (by first name) finished in last place most frequently in games played with the maximum number of players allowed? Exclude games designed for one player only.

Select the names from the multiple-choice options and enter your SQL query. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

- ☐ Barbara
- ☐ David
- ☐ Elizabeth
- ☐ James
- ☐ Jennifer
- ☐ John
- ☐ Linda
- ☐ Mary
- ☐ Michael
- ☐ Patricia
- ☐ Robert
- ☐ William

Question 19 & 20 (5 points)

What is the longest gap in days between two consecutive game nights?

Hint: Use DuckDB's date_diff() function.

Enter the result and your SQL query that produces exactly this result. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

Question 21 & 22 (5 points)

Which game(s) were played in a streak of three or more consecutive game nights?

Select the game titles from the multiple-choice options and enter your SQL query. SQL queries that output different or additional information will result in a loss of points. SQL queries that are unnecessarily

complex will also result in a loss of points. The SQL query must run in DuckDB. SQL queries that do not run in DuckDB will result in a loss of points.

- ☐ 7 Wonders
- ☐ Azul
- ☐ Carcassonne
- ☐ Catan
- ☐ Dixit
- ☐ Pandemic
- ☐ Splendor
- ☐ Ticket to Ride

JSON and JSON Pointer (10 points)

Solve the following tasks using the JSON document *prize.json* available on LearnIT.

Question 23 & 24 (2 points)

Create a JSON Pointer expression that accesses the year of the 500th prize in the JSON document *prize.json*.

Enter the result and the JSON Pointer expression that produces exactly this result.

Question 25 & 26 (2 points)

Create a JSON Pointer expression that accesses the last name (surname) of the only recipient of the 100th prize in the JSON document *prize.json*.

Enter the result and the JSON Pointer expression that produces exactly this result.

Question 27 & 28 (3 points)

Create a SQL query that answers how many prizes exist in the document. That is, count the length of the "prizes" array.

Enter the result and the SQL query that produces exactly this result.

Question 29 & 30 (3 points)

Create a SQL query that answers how many prizes with the category "peace" exist in the document.

Enter the result and the SQL query that produces exactly this result.

Role-Based Access Control (10 points)

Given five users (Barbara, David, Elizabeth, Jennifer, Robert) and three tables (players, boardgames, and gamenights), along with the SQL commands below, which define an RBAC setup following the Snowflake syntax as shown in Lecture 7, select all statements that apply in the following questions.

For completeness, note that all usage rights on databases, schemas, and warehouses exist for all roles and all users.

```
create role GameLead;
create role GameMaster;
create role Player;

grant role GameMaster to user Barbara;
grant role GameLead to user David;
grant role Player to user Jennifer;
grant role Player to user Robert;

grant role GameLead to role GameMaster;

grant select, delete, update on table gamenights to role GameMaster;
grant select, insert on table gamenights to role GameLead;
grant select on table players to role GameLead;
grant select on table players to role Player;
```

Question 31 (2 points)

Which users can insert new results in the gamenights table?

- ☐ Barbara
- ☐ David
- ☐ Elizabeth
- ☐ Jennifer
- ☐ Robert

Question 32 (2 points)

Which users can join the players and gamenights table to find who played some games already?

- ☐ Barbara
- ☐ David
- ☐ Elizabeth
- ☐ Jennifer
- ☐ Robert

Question 33 (2 points)

Which users can count the number of players in the players table?

- ☐ Barbara
- ☐ David
- ☐ Elizabeth
- ☐ Jennifer
- ☐ Robert

Question 34 (2 points)

Which users can delete results from the gamenights table in case they detect an error?

- ☐ Barbara
- ☐ David
- ☐ Elizabeth
- ☐ Jennifer
- ☐ Robert

Question 35 (2 points)

Which of the following statements about RBAC are correct?

- ☐ RBAC helps to limit the impact of prompt injection attacks.
- ☐ In RBAC there is always an account admin that can override settings, for example which users are granted which roles.
- ☐ RBAC is often used in large corporations because of its great scalability.
- ☐ RBAC is required in cloud-based database systems.

Indexes and Min/Max Summaries (10 points)

Assume a database stores the contents of the product table in the following three data files: Product1.data, Product2.data, and Product3.data. The database also stores metadata file containing min/max summaries: Minmax.metadata. Based on the min/max summaries and query predicates, the database applies pruning to reduce the number of files it needs to access. Assume the database does not cache any metadata or data and must always access files to read metadata and data for every SQL query.

The content of the files is as follows:

Product1.data

Header	
Small, 85	Tea, 100
Large, 100	Filter, 110

Product2.data

Header	
Fancy, 125	Espresso, 35
Cortado, 35	Mocha, 45

Product3.data

Header	
Jasmin, 30	Cola, 25

Minmax.metadata

Product1.data
Name: Filter – Tea
Price: 85 – 110

Product2.data
Name: Cortado – Mocha
Price: 35 – 125

Product3.data
Name: Cola – Jasmin
Price: 25 – 30

Question 36 (2 points)

Which files does the database minimally need to access to answer the following SQL query given the min/max summaries in Minmax.metadata?

```
select *
from product
where name = 'Espresso';
```

- ☐ Product1.data
- ☐ Product2.data
- ☐ Product3.data
- ☐ Minmax.metadata

Question 37 (2 points)

Which files does the database minimally need to access to answer the following SQL query given the min/max summaries in Minmax.metadata?

```
select *
from product
where name < 'Jasmin'
  and price > 35;
```

- ☐ Product1.data
- ☐ Product2.data
- ☐ Product3.data
- ☐ Minmax.metadata

Question 38 (2 points)

Which files does the database minimally need to access to answer the following SQL query given the min/max summaries in Minmax.metadata?

```
select min(price)
from product;
```

- ☐ Product1.data
- ☐ Product2.data
- ☐ Product3.data
- ☐ Minmax.metadata

Question 39 (2 points)

Which files does the database minimally need to access to answer the following SQL query given the min/max summaries in Minmax.metadata?

```
select avg(price)
from product;
```

- ☐ Product1.data
- ☐ Product2.data
- ☐ Product3.data
- ☐ Minmax.metadata

Question 40 (2 points)

Which of the following statements about joins and indexes are correct?

Select all that apply:

- ☐ The runtime performance of a hash join is linear.
- ☐ If data is already sorted, the database can utilize sort-merge joins efficiently.
- ☐ In databases, indexes typically summarize data to speed up aggregation functions like min(), avg(), and max().
- ☐ Without hash indexes, databases would be too slow to properly function.

BSc Only: Transactions (5 Points)

Only answer if you are a BSc student.

Select the correct answers:

- ☐ Atomicity ensures that all steps of a transaction are completed, or none are.
- ☐ Write-Ahead Logging (WAL) means that log records are written to disk before the corresponding data changes are written.
- ☐ Deadlocks occur when two transactions are waiting for each other to release locks, creating a cycle in the waits-for graph.
- ☐ The READ COMMITTED isolation level can lead to statements within a running transaction seeing committed changes of other transactions.
- ☐ Under the SERIALIZABLE isolation level only one user can log in to the database at any time.
- ☐ Using transactions ensures that data will never become corrupted.
- ☐ ACID properties guarantee that transactions can never be executed in parallel.
- ☐ Isolation means that a transaction does not see the effects of other transactions while it is running.
- ☐ ACID properties ensure that only authorized users can perform transactions.
- ☐ Transactions allow reading a consistent snapshot of data, even if the data needs to be read from different tables.

MSc Only: Big Data Management (5 points)

Only answer if you are an MSc student.

Select the correct answers:

- ☐ The Reduce phase in MapReduce aggregates the intermediate results produced by the Map phase.
- ☐ Graph databases are specifically designed for capturing and traversing relationships, such as in social networks.
- ☐ Key-value stores are great for storing multi-media objects such as music and videos.
- ☐ MapReduce is still widely used by Google for building web page indices as of 2025.
- ☐ Apache Hadoop is still the dominant framework for Big Data processing in 2025.
- ☐ Structured data is highly irregular and lacks repeating patterns, making it difficult to manage.
- ☐ Relational databases have historically been catching up in terms of performance with more specialized data-management systems, sometimes even rendering them obsolete.
- ☐ DuckDB is primarily designed for processing structured data and supports anti joins natively.
- ☐ DuckDB is optimized for in-memory analytical processing.
- ☐ ETL processes are only used in traditional relational databases, not in Big Data systems.